Problem 1a

Segment 3

$$\int_0^1 \frac{x^4+1}{x^6+1}\,\mathrm{d}x$$

Problem 1b

Segment 3

$$\int_0^1 \frac{x^2+1}{x^3+1}\,\mathrm{d}x$$

Problem 2a

Segment 3

$$\int_0^\infty \frac{1}{\sqrt{1+\sqrt{e^x}}}\,\mathrm{d}x$$

Problem 2b

Segment 3

$$\int_1^\infty e^{-\sqrt{\sqrt{x}-1}}\,\mathrm{d}x$$

Problem 3a

Segment 3

$$\int_0^\infty \frac{\sqrt[3]{x}}{1+x^3}\,\mathrm{d}x$$

Problem 3b

Segment 3

$$\int_0^\infty \frac{\sqrt[3]{x}}{1+x^4}\,\mathrm{d}x$$

Problem 4a

Segment 3

$$\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x^2}\right)}{x}\,\mathrm{d}x$$

Problem 4b

Segment 3

$$\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x$$

Problem 5a

Segment 3

$$\int_0^\infty \frac{\sqrt[3]{(1+\frac{3}{x})^x} - \sqrt{(1+\frac{2}{x})^x}}{x}\,\mathrm{d}x$$

Problem 5b

Segment 3

$$\int_{-\infty}^{\infty} \sin(x^2 + x\sqrt{3\pi} + \pi)\,\mathrm{d}x$$

Problem 6a

Segment 3

$$\int_0^{\frac{\pi}{4}} \cos^8(x) - \sin^8(x)\,\mathrm{d}x$$

Problem 6b

Segment 3

$$\int_0^{\frac{\pi}{4}} \cos^6(x) - \sin^6(x)\,\mathrm{d}x$$

Problem 7a

Segment 3

$$\int_0^1 \sqrt{\frac{x}{1-x}} \ln\left(\frac{x}{1-x}\right) \, \mathrm{d}x$$

Problem 7b

Segment 3

$$\int_0^1 \frac{\ln(\sqrt{x} + \sqrt{1-x})}{\sqrt{x}} \, \mathrm{d}x$$